Lecture 5. Movement of Labor and Capital between Countries

Lectures on International Trade

Instructor: Xiaoping Chen

Harbin Institute of Technology, Shenzhen

Recall from Previous Lectures

- Goods and services are traded, factors can not move across countries.
- However, there are international flows of factors of production.
  - Movement of labor and capital between countries: Migration and Foreign direct investment.
  - Foreign aids and loans.
  - Natural disasters, wars may destroy factors of production.
- What are the effects of these movements?

Outline

- Movement of Labor between Countries: Migration
- Movement of Capital between Countries: Foreign Direct Investment
- Gains from Labor and Capital Flows

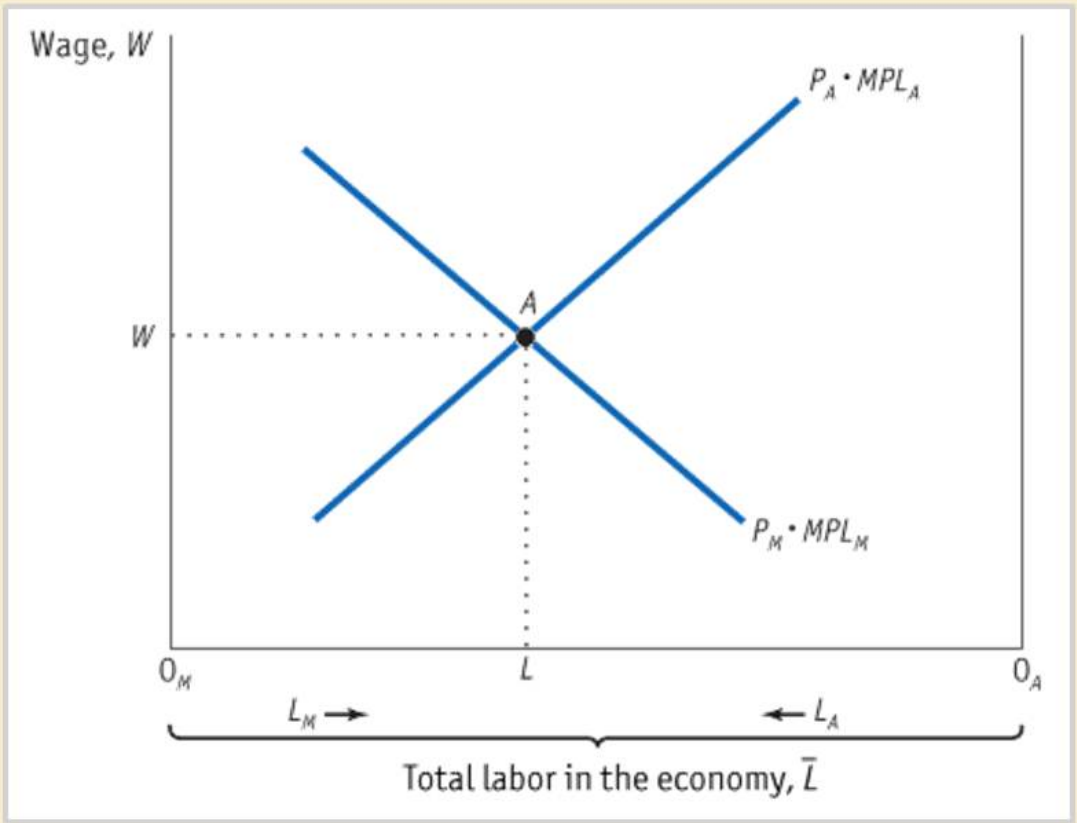
Outline

- Movement of Labor between Countries: Migration
- Movement of Capital between Countries: Foreign Direct Investment
- Gains from Labor and Capital Flows
- Short-run analysis: Specific-factor model
- Long-run analysis: Heckscher-Ohlin model

Migration

Effects of Immigration in the Short Run: Specific-Factors Model

FIGURE 5-1 Home Labor Market



$$L_M + L_A = \bar{L}$$

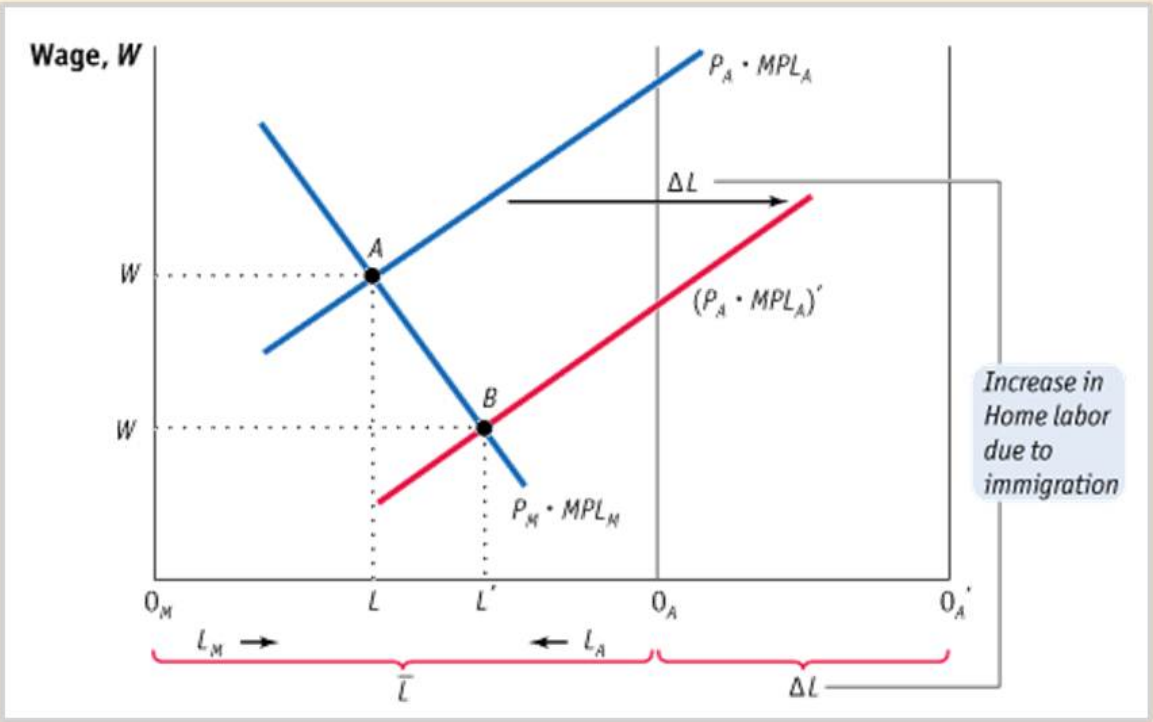
The Home wage is determined at point *A*, the intersection of the marginal product of labor curves $P_M \cdot MPL_M$ and $P_A \cdot MPL_A$ in manufacturing and agriculture, respectively.

At point *A*, $0_M L$ units of labor are used in manufacturing and $0_A L$ units of labor are used in agriculture.

Migration, cont'

Effects of Immigration on Wage in Home:

FIGURE 5-2 Increase in Home Labor



When the amount of labor at Home increases by the amount ΔL, the origin for agriculture shifts to the right by that amount, from 0_A *to* $0A'$.

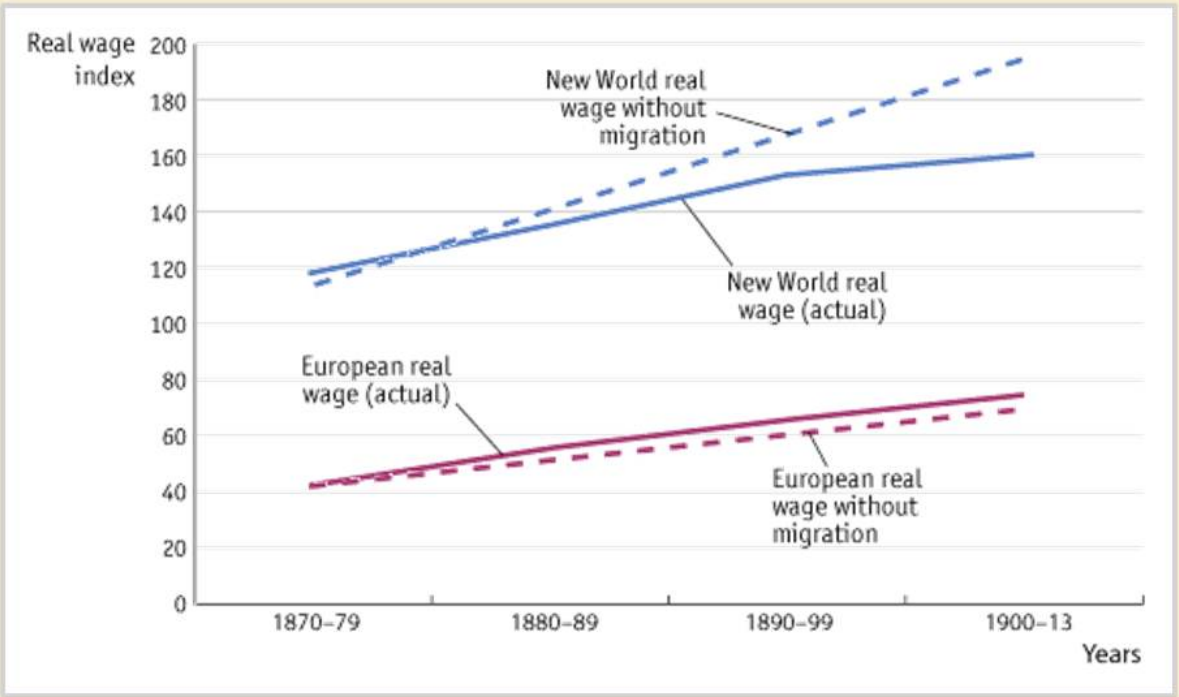
The marginal product of labor curve in agriculture also shifts right by the amount ΔL.

Equilibrium in the Home labor market is now at point *B:* wages have fallen to W' and the amount of labor has increased in manufacturing (to $0_M L'$) and in agriculture (to $0_A L'$).

Migration, cont'

Effects of Immigration to the New World: An Application

FIGURE 5-3 **Wages in Europe and the New World**



Large-scale migration from Europe to the New World in America and Australia closed the wage gap between the two locations. In 1870 wages in the New World were almost three times as high as wages in Europe, whereas in 1910 they were about twice as high.

Migration, cont'

Effects on the Rentals of Capital and Land in the Short-Run:

- U.S. and Europe have both welcomed foreign workers in specific industries: agriculture and high-tech.
- They do this even though those foreign workers compete with domestic workers in those industries.
- As more labor is hired in each industry (because wages are lower), the marginal products of capital and land both increase. The increase in the marginal product occurs because each machine or acre of land has more workers available to it, and that machine or acre of land is therefore more productive.
- So the marginal products of capital and land rise and so do their rentals.

Migration, cont'

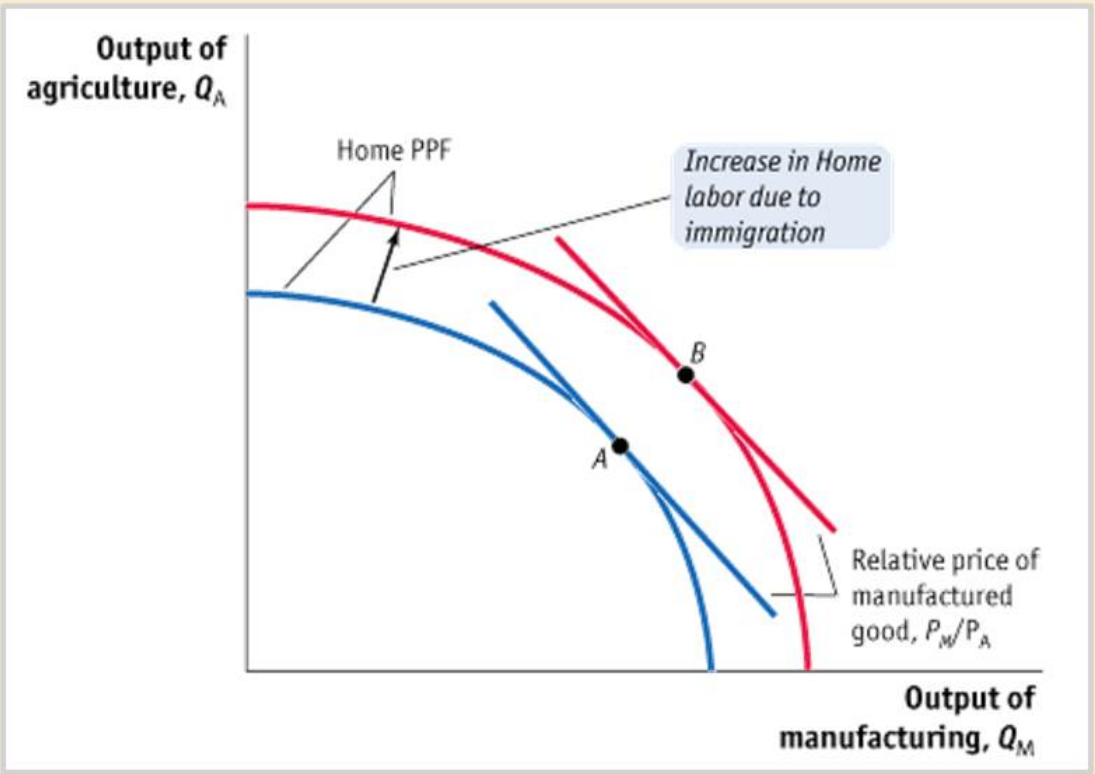
Effects on the Rentals of Capital and Land in the Short-Run:

- From this line of reasoning, we should not be surprised that owners of capital and land often support more open borders, which provides them with foreign workers that can be employed in their industries.
- The restriction on immigration in a country should therefore be seen as a compromise between entrepreneurs, landowners who might welcome the foreign labor and local unions and workers who view migrants as a potential source of competition leading to lower wages.
- Immigrant groups themselves, if they are large enough, might also have the ability to influence the political outcome on immigration policy.

Migration, cont'

Effect of Immigration on Industry Output in the Short-Run:

FIGURE 5-5 Shift in Home Production Possibilities Curve



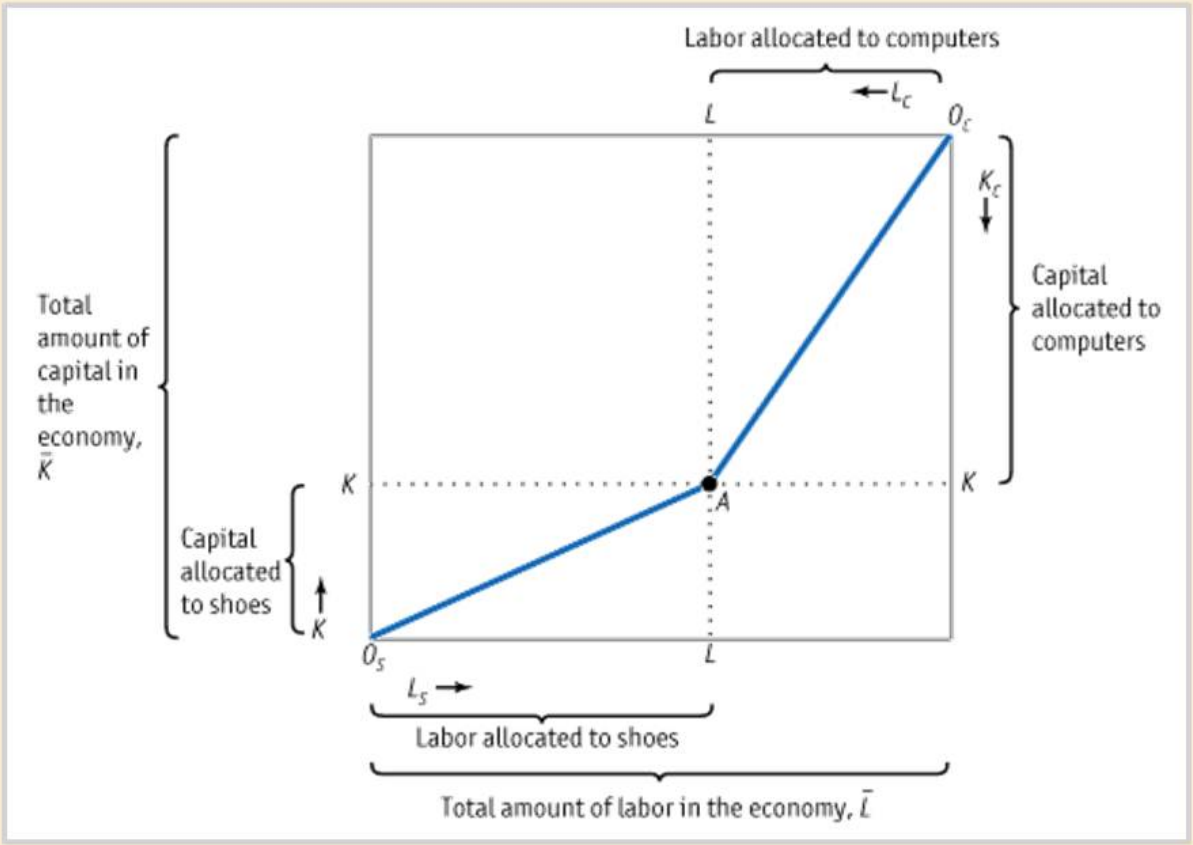
With the increase in labor at Home from immigration, the production possibilities frontier shifts outward and the output of both industries increases, from point *A* to point *B*.

In the short run, land and capital do not move between the industries, and the extra labor in the economy is shared between both industries.

Migration, cont'

Effect of Immigration on Industry Output in the Long-Run: The K/L ratios in the two industries are measured by the slopes of $O_S A$ and $O_C A$.

FIGURE 5-7 Allocation of Labor and Capital in a Box Diagram



The top and bottom axes of the box diagram measure the amount of labor, L, in the economy, and the side axes measure the amount of capital, K.

At point A, $0_S L$ units of labor and $0_S K$ units of capital are used in shoe production, and $0_C L$ units of labor and $0_C K$ units of capital are used in computers.

Migration, cont'

Effects of Immigration in the Long Run:

- To determine the wage and rental in the economy, we use the marginal products of labor and capital, which are in turn determined by the capital-labor ratio in either industry.
- If there is a higher capital-labor ratio (that is, if there are more machines per worker), then by the law of diminishing returns, the marginal product of capital and the real rental must be lower.
- Having more machines per worker means that the marginal product of labor (and hence the real wage) is higher because each worker is more productive.

Migration, cont'

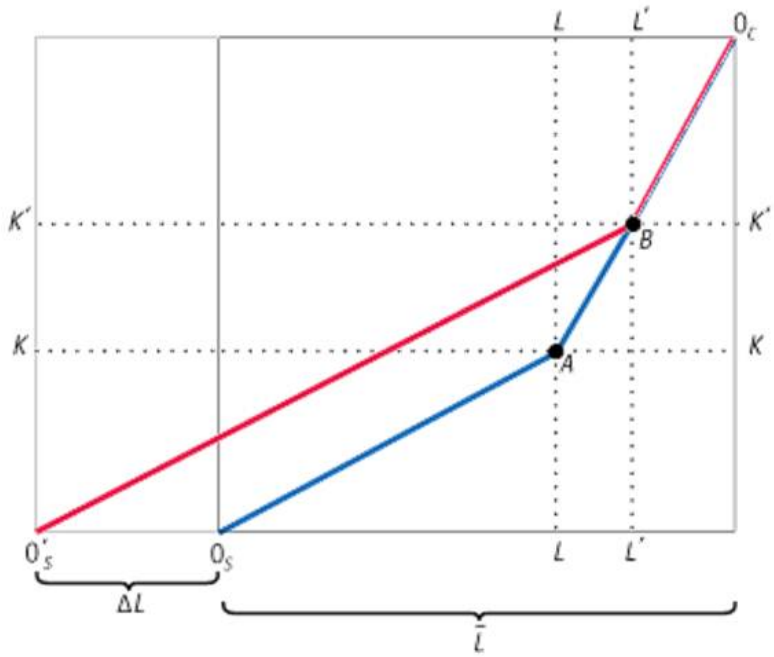
Effects of Immigration in the Long Run: An increase of Home labor

- Suppose Home labor increase due to immigration,
- How will these additional labor be allocated?
  - Some in the shoe industry, some in the computer industry?
  - Will capital also move because of this change of labor?
- The answer is, different from the short-run specific-factor model, all the additional labor will be allocated in the labor-intensive show industry.
- Even more labor and capital will move from the computer industry to the show industry.

Migration, cont'

Effects of Immigration in the Long Run: An increase of Home labor

FIGURE 5-8 (1 of 2) **Increase in Home Labor**



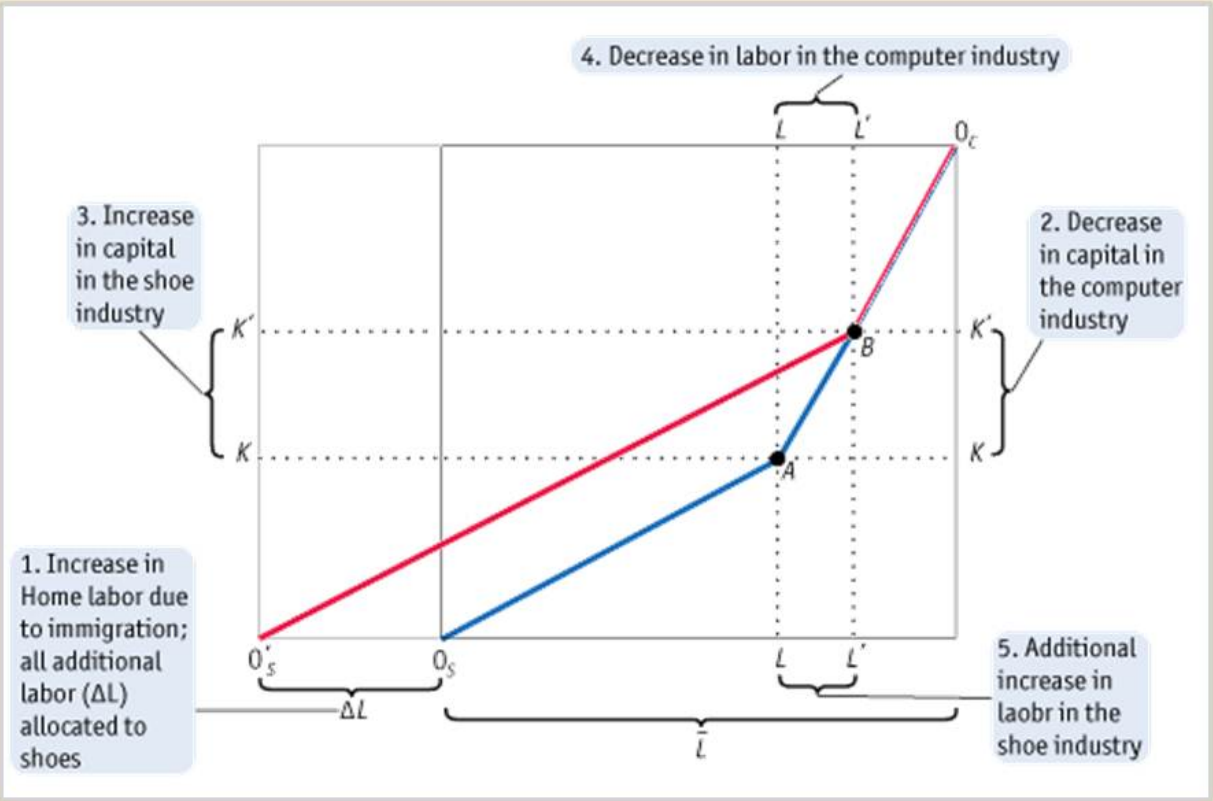
With an increase in Home labor by ΔL, the origin for the shoe industry shifts from 0_S *to* $0_S'$.

At point *B*, $0'_S L'$ units of labor and $0'_S K'$ units of capital are used in shoes, while $0_C L'$ units of labor and $0_C K'$ units of capital are used in computers.

Migration, cont'

Effects of Immigration in the Long Run: An increase of Home labor

FIGURE 5-8 (2 of 2) **Increase in Home Labor (continued)**

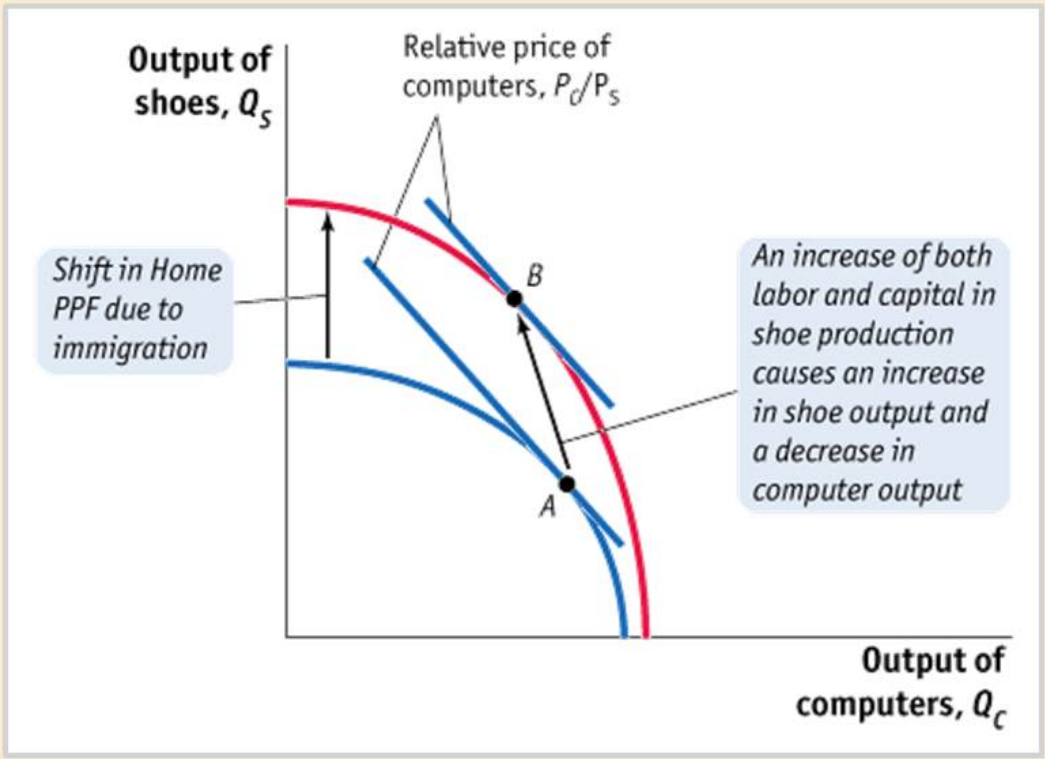


In the long run, industry outputs adjust so that the capital-labor ratios in each industry at point B (the slopes of $0_S'B$ and 0_CB) are unchanged from the initial equilibrium at point A (the slopes of 0_SA and 0_CA). All new labor from immigration is allocated to the shoe industry, *additional* capital and labor are transferred from computers to shoes, keeping the capital-labor ratio in both industries unchanged.

Migration, cont'

Effects of Immigration in the Long Run Industry Output: An increase of Home labor

FIGURE 5-9 The Long-Run Effect on Industry Outputs of an Increase in Home Labor



With an increase in the amount of labor at Home, the PPF shifts outward.

The output of shoes increases while the output of computers declines as the equilibrium moves from point *A* to *B*.

The prices of goods have not changed, so the slopes of the PPFs at points *A* and *B* (i.e., the relative price of computers) are equal.

Long-Run Factor Endowment Change

Rybczynski Theorem

- The finding that an increase in labor will expand one industry but contract the other holds only in the long run; in the short run, as we saw in Figure 5-5, both industries will expand.
- The **Rybczynski theorem** states that, in the Heckscher-Ohlin model with two goods and two factors, an increase in the amount of a factor found in an economy will increase the output of the industry using that factor intensively and decrease the output of the other industry.
- We have proved the Rybczynski theorem for the case of immigration, where labor in the economy grows.
- As we argue later in this lecture, the same theorem holds when capital in the economy grows: in this case, the industry using capital intensively expands and the other industry contracts.

Long-Run Effect on Factor Prices

Recall that the wage-rental ratio doesn't change due to factor endowment change in the long-run.

- Factor prices do not need to change as a result of immigration.
- The reason that factor prices do not need to change is that the economy can absorb the extra amount of a factor by increasing the output of the industry using that factor intensively and reducing the output of the other industry.
- The finding that factor prices do not change is sometimes called the **factor price insensitivity** result.

International Movement of Capital

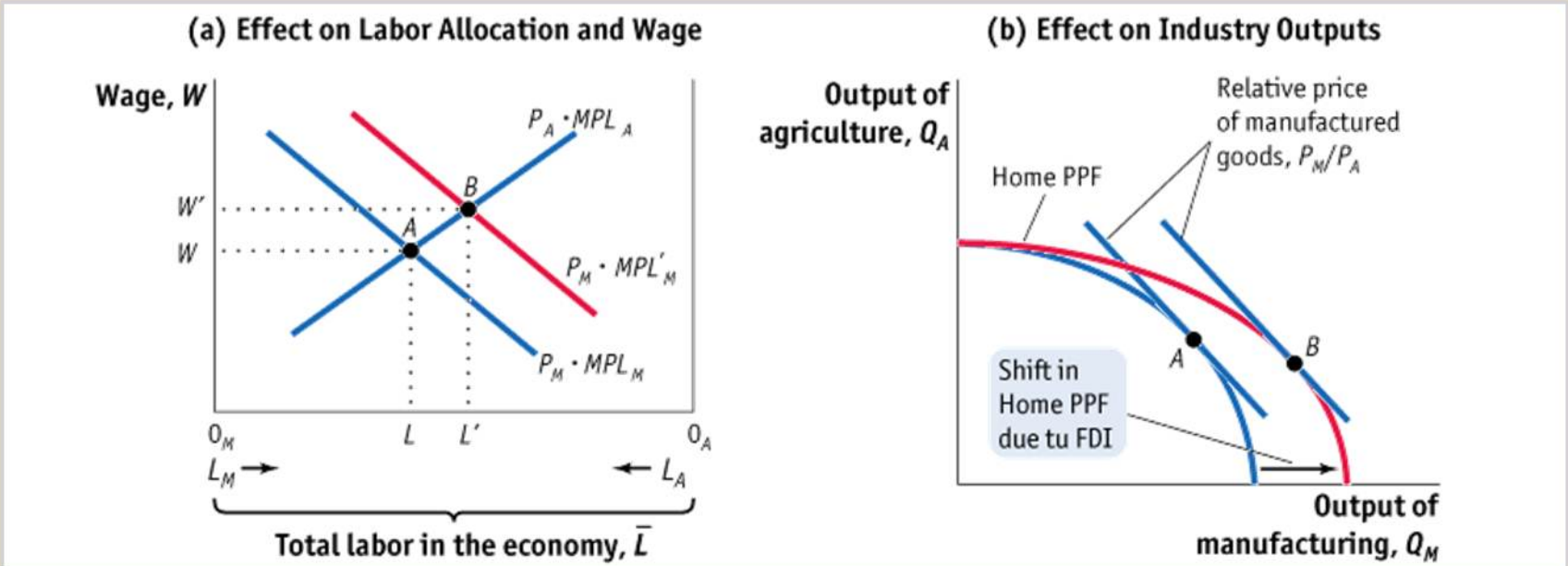
We now look at how capital can move from one country to another through **foreign direct investment (FDI)**, which occurs when a firm from one country owns a company in another country.

- Our focus in this section will be on Greenfield investment, that is, the building of new plants abroad.
- We model FDI as a movement of capital between countries, just as we modeled the movement of labor between countries.
- The key question we ask is: How does the movement of capital into a country affect the earnings of labor and capital there?
- This question is similar to the one we asked for immigration, so the earlier graphs that we developed can be modified to address FDI.

Foreign Direct Investment, cont'

FDI in the Short-Run: Specific-Factors Model

FIGURE 5-11 Increase in the Capital Stock in the Short Run



In panel (a), an inflow of capital into the manufacturing sector shifts out the marginal product of labor curve in that sector. In panel (b), with the inflow of capital into manufacturing, and the extra labor used in that sector, the output of manufacturing increases.

Because labor has been drawn out of agriculture, the output of that sector falls. These changes in outputs are shown by the outward shift of the PPF (due to the increase in capital) and the movement from point *A* to point *B*.

Foreign Direct Investment, cont'

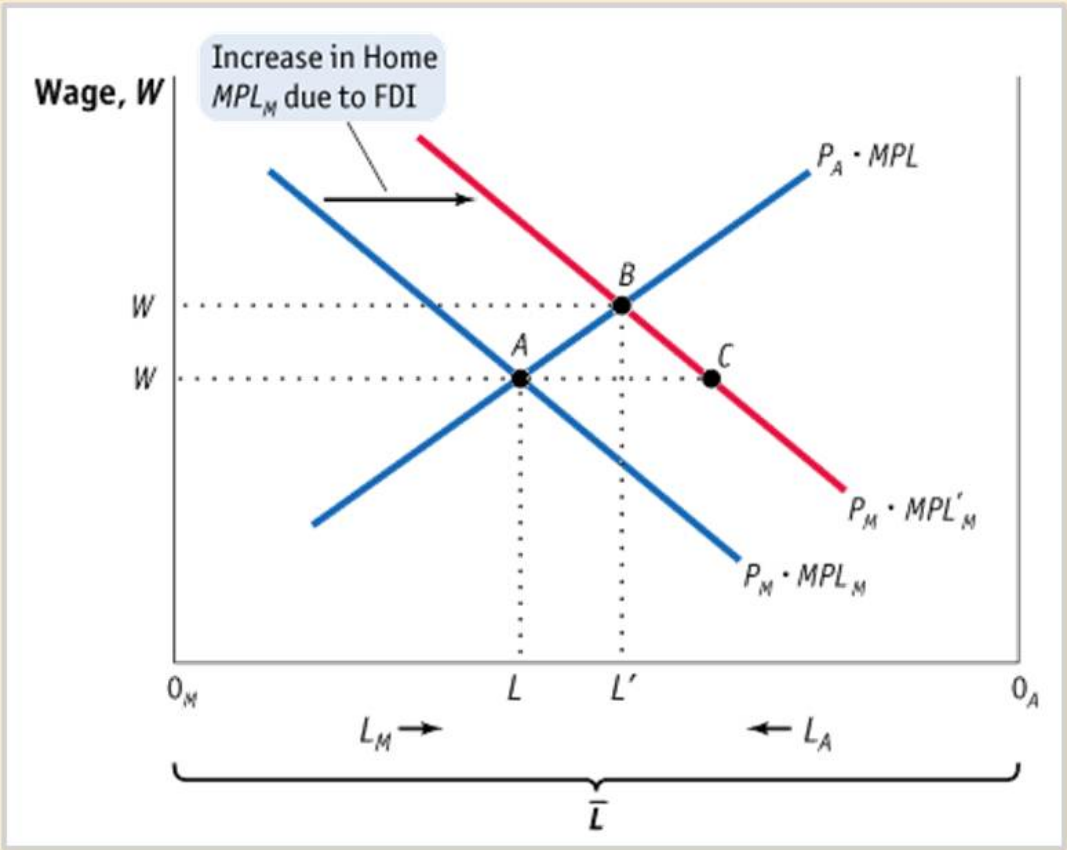
FDI in the Short-Run: Effect of FDI on the Rentals

- With regard to the rental on land, we know that with an inflow of FDI, fewer workers are employed in agriculture, and each acre of land cannot be used as intensively.
- The value of marginal product of land, $R_L = P_A \cdot MPT_A$, falls. Because when MPT_A falls and P_A remains unchanged, then land real rental must fall.
- How about the real rental to capital?

Foreign Direct Investment, cont'

FDI in the Short-Run: Effect of FDI on the Rentals

FIGURE 5-12 The Effect of an Increase in Capital Stock on the Rental on Capital



By carefully tracing through how the capital-labor ratio in manufacturing is affected by the movement from *A* to *C* (where wages and hence the capital-labor ratio do not change), and then the movement from *C* to *B* (where wages and the capital-labor ratio both increase), we conclude that the rental on capital is lower at point *B* than at point *A*. Therefore, the rental on capital declines when the capital stock increases through FDI.

Foreign Direct Investment, cont'

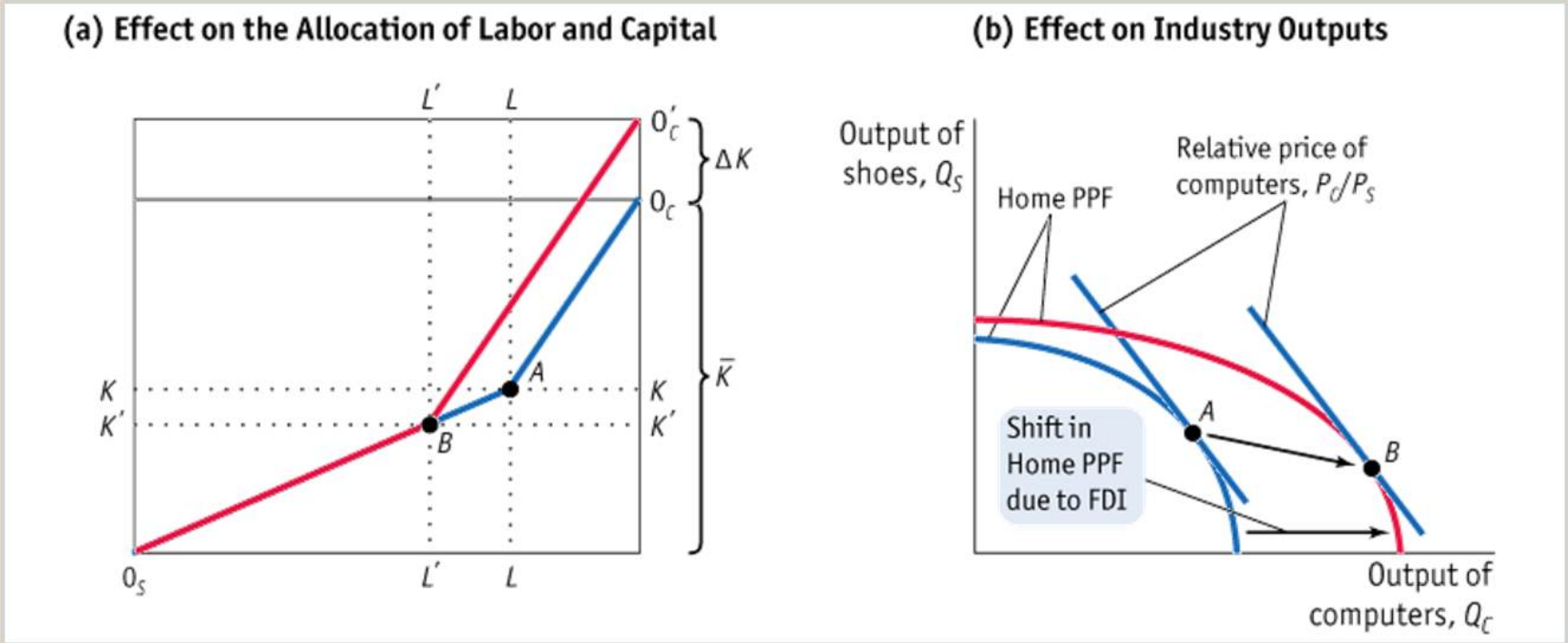
FDI in the Long-Run

- To analyze the impact of FDI in the long run, we continue with the same assumptions as before:
  - There are two industries, computers and shoes, with two factors, labor and capital.
  - Computers are capital intensive and shoes are labor intensive.

Foreign Direct Investment, cont'

FDI in the Long-Run

FIGURE 5-13 Increase in the Capital Stock in the Long Run



In panel (a) the horizontal axes of the box diagram measure the amount of labor in the economy, and the vertical axes measure the amount of capital. Home capital increases by ΔK.

In panel (b), with the increase in the amount of capital at Home from increased FDI, the PPF shifts outward. More for the computer output.

Foreign Direct Investment, cont'

FDI in the Long-Run on Output and Factor Prices

- Capital increases due to FDI.
- As the Rybczynski Theorem states, the increase in capital through FDI has increased the output of the capital-intensive industry and reduced the output of the labor-intensive industry.
- This change in output is achieved with no change in the capital labor ratios in either industry.
- Because capital-labor ratios are unchanged, the wage and the rental on capital are also unchanged.
- In the long run model, an inflow of either factor of production will leave factor prices unchanged.

Gains from Labor and Capital Flows

Policy on Immigration and FDI

- Foreign investment and immigration are both controversial policy issues.
- Most countries have at some point controlled FDI but later became open to foreign investment.
- However, almost all countries impose limits on immigration. U.S. immigration controls were established by the Quota Law of 1921. It allows a limited number of persons arriving annually from each country of origin.
- The Immigration and Nationality Act Amendments of 1965 revised the country-specific limits and allowed immigration on a first-come first-served basis up to a limit.
- Subsequent revisions to the immigration laws in the U.S.

Gains from Labor and Capital Flows, cont'

Why is immigration so controversial?

- Some groups oppose the spending of public funds on immigration.
- Other groups fear the competition for jobs created by an inflow of workers.
- Does immigration provide an overall gain to the host country, not including the gains to the immigrants themselves?
- Are there overall gains to the destination country, in the same way as we have found overall gain from trade?

Gains from Labor and Capital Flows, cont'

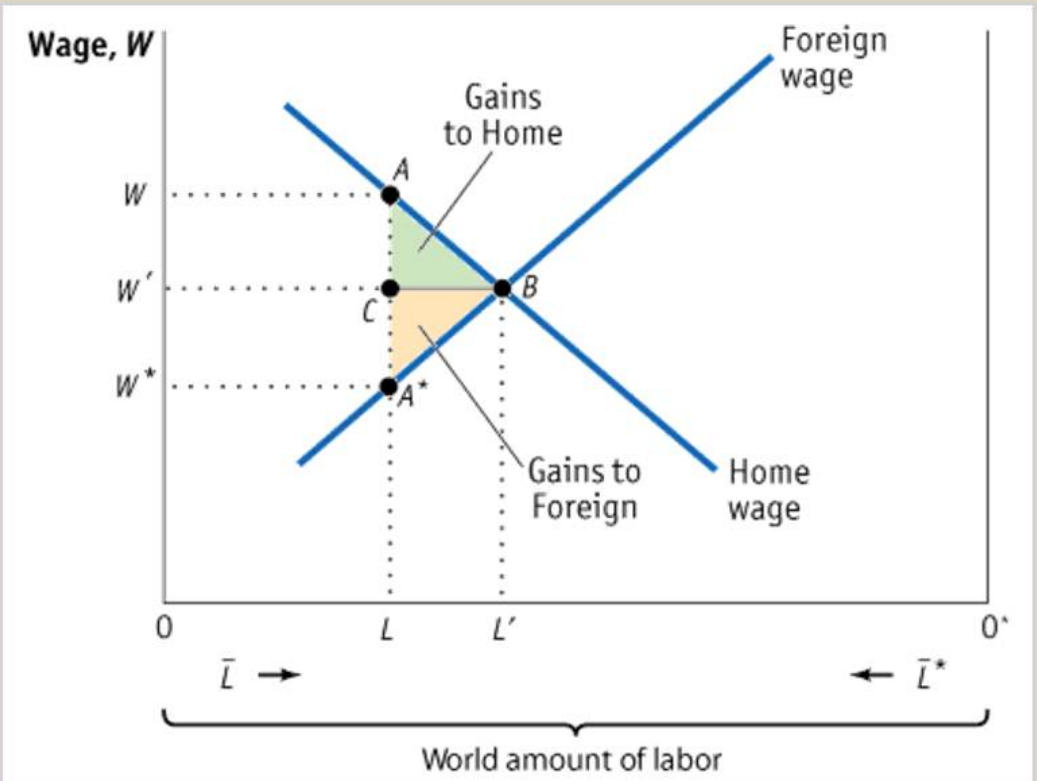
Gains from Immigration and FDI

- Immigration benefits the host country in the specific factors model.
- If we include the immigrant earnings with Foreign income, then we find that emigration benefits the Foreign country, too.
- The same argument can be made for FDI.
- An inflow of capital benefits the host country not including the extra earning of foreign capital.
- If we count those extra earnings then FDI also benefits the source country for the capital.

Gains from Labor and Capital Flows, cont'

Gains from Immigration and FDI

FIGURE 5-14 World Labor Market



Initially, Home has OL workers and Foreign has O^*L workers.

The Home wage is W, as determined at point A, which is higher than the Foreign wage W^* at A^*.

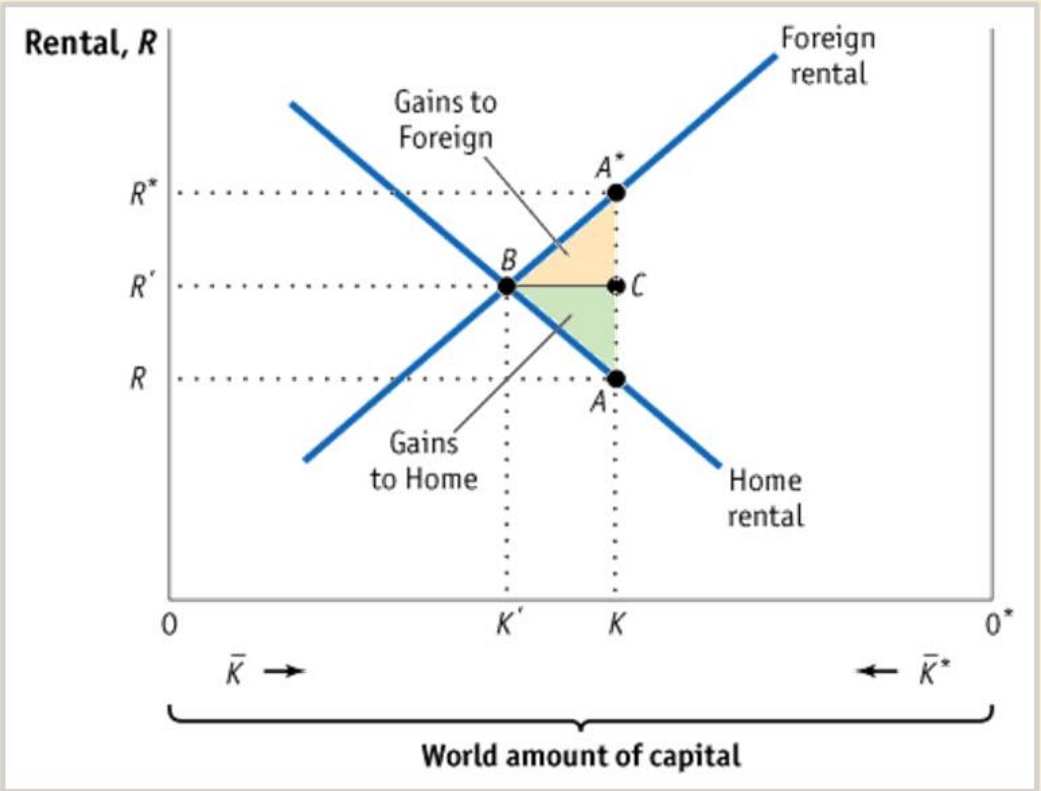
Workers will move from Foreign to Home to receive higher wages. The equilibrium with full migration is at point C, where wages are equalized at W'.

The gain to Home from migration is measured by triangle ABC, and triangle A^*BC represents the gains to Foreign.

Gains from Labor and Capital Flows, cont'

Gains from Immigration and FDI

FIGURE 5-15 World Capital Market



With $0K$ units of capital at Home, the Home rental is R, at point A.

The remaining capital 0^*K is in Foreign, and the Foreign rental is R^*, at point A^*.

Capital will move from Home to Foreign to receive a higher rental. The equilibrium with full capital flows is at point B, where rentals are equalized at R'.

Triangle ABC measures the gains to Home from the capital outflow, and triangle A^*BC measures the gains to Foreign.

Summary

Key Points:

1. Holding the amount of capital and land fixed in both industries, as in the specific-factors model, immigration leads to a fall in wages. This was the case, for example, with the mass migration to the New World in the nineteenth century.

2. As wages fall due to immigration, the marginal products of the specific factors (capital and land) rise, and therefore their rentals also increase.

3. Fixing the amount of capital and land in a country is a reasonable assumption in the short run, but in the longer run, firms will move capital between industries, which will change the effect of immigration on wages and rentals.

Summary, cont'

4. In a long-run model with two goods and two factors, both of which are perfectly mobile between the industries, additional labor from immigration will be absorbed entirely by the labor-intensive industry. Furthermore, the labor-intensive industry will also absorb additional capital and labor from the capital-intensive industry, so its capital–labor ratio does not change in the long run. Because the capital–labor ratio in each industry does not change, the wage and rentals remain the same as well. This results in what is known as factor price insensitivity.

Summary, cont'

5. According to the Rybczynski theorem, immigration will lead to an increase in output in the labor-intensive industry and a decrease in the output of the capital-intensive industry. This result is different from that of the short-run specific-factors model, in which immigration leads to increased output in both industries.

6. Besides trade in goods and the movement of labor, another way that countries interact with one another is through investment. When a company owns property, plant, or equipment in another country, it is called foreign direct investment, or FDI.

7. In the short run, FDI lowers the rentals on capital and land and raises wages. In the long run, the extra capital can be absorbed in the capital-intensive industry without any change in the wage or rental.

Summary, cont'

8. According to the Rybczynski theorem, FDI will lead to an increase in the output of the capital-intensive industry and a decrease in the output of the labor-intensive industry.

9. The movement of capital and labor generates overall gains for both the source and host countries, provided that the income of the emigrants is included in the source country's welfare. Hence, there are global gains from immigration and FDI.